

EXPERIENCE

JAGCO | *Software Engineer Intern*

8/2024 - present

- Developed a **reception AI system** using **Python** and **OpenAI's LLMs** to handle real-time, multi-turn conversations with office visitors; enabled facial recognition-based identification with 95% accuracy and reduced front desk wait times by 70%.
- Engineered an **automation solution** for the **Jaka cobot**, integrating **python-based** voice command functionality to facilitate task execution through spoken instructions.
- **Built an AI-powered proposal aggregation app** that scrapes and summarizes up to **1,200+** active government solicitations from SAM.gov, Grants.gov and DSIP weekly; users can also generate response drafts using fine-tuned LLMs, accelerating draft creation time by **80%**.
- Enhanced semantic search using **LangChain**, **LlamaIndex**, boosting retrieval accuracy by **30%**.
- Delivered demos to 10+ non-technical stakeholders, driving faster buy-in through simplified AI workflow explanations.

BRANE ENTERPRISES PVT. LTD | *Software Developer*

6/2021 - 6/2023

- Developed a **no-code/low-code app** using **Java** and **Spring Boot**, reducing dev time by 40% and improving release agility.
- Integrated **RESTful APIs**, increasing data flow efficiency by **30%** and reducing manual intervention.
- Delivered full-stack features in **agile** sprints, collaborating across global teams to meet sprint goals on time.
- Developed responsive UIs with **React.js** and automated deployments via **Docker and CI/CD**, accelerating release cycles by **50%**.
- Achieved **90%+** test coverage with **JUnit**, improving code maintainability and reducing bugs in production.
- Developed and maintained scalable web applications using **JavaScript** with frameworks like **React** and **Angular**, ensuring responsive and user-friendly interfaces.
- Worked in **Linux/Unix-based environments**, managing file systems, scripting, and application deployment.
- Participated in the design, deployment, and monitoring of applications on **Kubernetes** and cloud infrastructure, utilizing **Terraform**.
- Built responsive user interfaces with **Angular**, integrating seamlessly with **RESTful APIs** to enhance user experience.
- Integrated Git and Jenkins into the **CI/CD** pipeline to automate build, test, and deployment processes, improving development efficiency.
- Collaborated in cross-functional **Agile** teams, participating in daily stand-ups, sprint planning, and retrospectives to deliver iterative improvements.
- Wrote complex **SQL** queries and optimized database performance to support robust backend functionalities.
- Communicated technical concepts and project progress effectively with stakeholders and team members, both in writing and verbally.

RESEARCH & TEACHING EXPERIENCE

ARIZONA STATE UNIVERSITY | *Graduate Research Assistant in Robotics*

10/2023 - present

- Engineered a **large language model-based task decomposition framework** that translated high-level instructions into precise, executable actions for Cobots, enabling autonomous multi-step task execution.
- Developed **kinematic-aware prompting** strategies using domain-specific constraints and integrated them with **LangChain**, enhancing robotic control precision and reliability.
- Integrated **LLM-driven reasoning** into robotic systems using **ROS**, facilitating adaptive automation and reducing dependency on static, pre-programmed routines.

- ARIZONA STATE UNIVERSITY | *Graduate Teaching Assistant*01/2024 - present
- Supported instruction and assessment for **300+** students across a range of engineering and computing courses, including Software Engineering (CSE 360), Thermo-Fluids (EGR 340), Robotic Systems I & II (RAS 545/546), and Systems Administration (IFT 220).
 - Facilitated review sessions, provided individualized academic guidance, and ensured consistent, equitable evaluation to enhance student learning outcomes.

PROJECTS

- E-COMMERCE PLATFORM WITH REAL-TIME CHAT & RECOMMENDATION SYSTEM
- Developed **RESTful APIs** and real-time chat using **WebSockets**.
 - Designed responsive UIs with **React/Redux**, ensuring efficient frontend architecture.
 - Developed **data analysis workflows** in **Python**, utilizing **Pandas** for data manipulation and **Matplotlib** for visualization.
- SECUREPIX – SECURE IMAGE ENCRYPTION WEB APPLICATION
- Developed a full-stack web application for secure image encryption and decryption using **AES (Advanced Encryption Standard)** in ECB and CBC modes, with robust key handling via **PBKDF2 and HMAC** integrity checks.
 - Engineered the backend using **Python, Django, and PyCryptodome**, with frontend interactivity via **HTML/CSS, JavaScript, and Tkinter GUI** for desktop prototype.

TECHNICAL SKILLS

- Languages:** Python, Java, JavaScript, SQL
- Front-End:**, React, HTML, CSS
- Machine Learning Frameworks:** HuggingFace Tensorflow, Keras, PyTorch, GPT, OpenCV, NLTK
- Databases:** PostgreSQL, MongoDB, OracleDB, MySQL
- Tools:** Git, Jenkins, Postman, Maven, Jupyter Notebook, Confluence, Jira, Kubernetes, Agile Methodologies, AWS
- Operating System:** Linux/Unix, Windows, MacOS
- Visualization Tools:** Tableau, PowerBI, Matplotlib, Plotly, Bokeh

EDUCATION

ARIZONA STATE UNIVERSITYTempe, AZ
Masters of Science in Information Technology / *CGPA: 4.33/4.0*2023 - present